EVOCURE Candidate Screening

Pancreatic ductal adenocarcinoma, five prioritised compounds

EVOCURE-PDAC-20261027 | generated 2026-10-27 22:29:42 UTC

What this document is

A demonstration of an open source in silico screening platform. It shows what the software computed on one reproducible run: a bounded evolutionary search over molecular structures, scored by a QSAR model, toxicology heuristics and a simulated two dimensional tumour microenvironment.

What it is not: evidence that any compound here treats pancreatic cancer. No molecule in this report has been synthesised, assayed in a cell, or given to an animal or a person. These are computational hypotheses at the earliest possible stage of discovery, and the large majority of such hypotheses fail.

Result in one line

No candidate of 5 separated from the negative control, and neither did the positive control when scored by the same potency model.

The screening pipeline ran end to end and produced five ranked candidates. The tissue assay was checked with a positive and a negative control before the candidates were read, so the result above is interpretable rather than merely generated. The positive control separates from the negative control, so the tissue model responds to a known active in this configuration.

Run at a glance

Structures evaluated	109 unique valid structures
Generations	4, population 24
Replicate seeds per compound	5
Compute backend	NumPy CPU float64
QSAR training set	43 curated compounds
QSAR scaffold split R2	0.699
QSAR classifier ROC AUC	0.954
Wall clock	309 seconds

Open source platform and data. Reproduce this document with the commands in the reproducibility section. Report generated directly from the run's results file; no figure in it was entered by hand.

1. Summary

A bounded evolutionary search started from four approved pancreatic cancer drugs and generated structural variants through point mutations and scaffold recombination. Every variant was validated with RDKit, scored on five objectives, and the non dominated front was carried into the next generation. The five compounds below are the highest ranked variants that were not themselves seeds.

1.1 Assay validity, checked before reading the candidates

Role	Compound	TGI %	95% CI	CI excludes 0
Positive, curated	Gemcitabine	67.1	60.4 to 73.8	yes
Negative	Caffeine	-5.7	-18.3 to 6.8	no
Positive, modelled	Gemcitabine	-1.0	-14.7 to 12.7	no

The third arm is the same positive control molecule with its curated bioactivity withheld, so its potency comes from the structure-property model that scores every generated candidate. It is the only arm the candidates can fairly be compared against.

Does the simulated assay work?
Positive versus negative control: $p < 0.001$, mean difference 72.9 percentage points, Hedges g 8.12.

The positive control separates from the negative control, so the tissue model responds to a known active in this configuration.

1.1a The comparison is between a measurement and a prediction

The positive control fails its own test when scored like a candidate
Gemcitabine draws its potency from curated experimental bioactivity: IC50 0.0450 uM, and it inhibits growth by 67.1 %. A generated structure has no measurement, so it is scored by the structure-property model. Scoring this same molecule that way gives IC50 8.053 uM, 179 times weaker, and growth inhibition falls to -1.0 %, which does not separate from the negative control.

Nothing about the molecule changed. Only the provenance of its potency did. Any candidate in this report is therefore being compared against a number no generated structure can ever be assigned, and failing that comparison says little about the chemistry.

1.2 The five candidates

ID	Compound	pIC50	Score	TGI %	95% CI	vs control
Candidate 01	Compound-CFB9B6	7.03	0.572	9.4	-6.8 to 25.5	not distinguishable
Candidate 02	Compound-5D822D	7.13	0.564	5.5	-6.8 to 17.9	not distinguishable
Candidate 03	Compound-FAFE93	5.97	0.544	1.6	-8.9 to 12.2	not distinguishable
Candidate 04	Compound-C5F9E6	6.97	0.542	5.2	-5.3 to 15.6	not distinguishable
Candidate 05	Compound-32914B	7.10	0.540	0.5	-13.5 to 14.4	not distinguishable

TGI is tumour growth inhibition in the simulated microenvironment, relative to an untreated arm, averaged over independent seeds. Score is the weighted composite across the five search objectives. The last column reports Welch's t-test of each candidate against the negative control; it is the column that decides whether a point estimate means anything.

Read this before using the ranking

No candidate separated from the negative control in the tissue simulation. The ranking above is therefore driven by predicted properties, drug likeness, toxicity alerts and applicability domain, not by a demonstrated simulated effect.

This is not evidence that the molecules are inactive. The positive control fails the identical test when scored by the same potency model, so the test does not currently distinguish a good candidate from a known drug. What the run establishes is a limit of the platform, not a property of these five structures.

The honest reading is that the search found chemically reasonable, low alert, drug like molecules near the seed scaffolds, and that this experiment cannot say whether any of them is active. Treat the list as a starting set for further computational work, not as a shortlist of likely treatments.

2. Method

2.1 Search

Objective prompt	Find 5 experimental candidates for pancreatic cancer based on Gemcitabine, 5-Fluorouracil, Erlotinib and Paclitaxel. Prioritize activity, low predicted toxicity and drug-likeness.
Planner	local_rule_based_planner
Derived priorities	safety, druglikeness, activity
Seed compounds	Gemcitabine, 5-Fluorouracil, Paclitaxel, Erlotinib
Generations	4
Population size	24
Applied concentration	5.0 uM
Exposure duration	24.0 hours
Search seed	20261027

The planner converts a plain language objective into search priorities only. It does not propose molecules and it does not supply any activity, potency or safety number. All quantitative output comes from the deterministic scientific models.

2.2 Objective weights

Objective	Weight	Source of the value
Predicted PDAC activity	2.0	Calibrated QSAR regressor, sigmoidal utility at pIC50 6.0
Low toxicological risk	2.0	Ten category structural alert and rule based risk index
Drug likeness (QED)	2.0	RDKit quantitative estimate of drug likeness
Selectivity index	1.0	ADMET derived tumour versus healthy cell window
Applicability domain	1.0	Maximum Tanimoto similarity to the training set
Simulated growth inhibition	1.0	Agent based tumour microenvironment, treated versus untreated

Tumour cell kill is never maximised alone. A compound that scores well on activity but triggers toxicity alerts or falls outside the applicability domain is penalised by construction.

2.2a What selection actually optimises

The selection loop is closed

Every candidate that could be dosed was simulated in the tumour microenvironment, 178 runs in total, and its growth inhibition entered the objective set used for Pareto sorting and survivor selection. The tumour model therefore exerts selective pressure on the population rather than only reporting on the winners.

Two limits on how much that is worth. All arms share one seed per generation, so the comparison is paired but the objective is the noisiest of the six. And the tissue model's drug response is driven by the predicted IC50, so this objective is correlated with the activity objective rather than independent of it. It adds diffusion, uptake, hypoxia and clonal structure on top of that prediction; it does not replace it with a measurement.

2.3 The QSAR model, and what it can carry

Training compounds	43
Independent scaffolds	32
Validation	repeated scaffold-grouped CV, 5 folds x 10 repeats
Pooled test predictions	430
Regression RMSE	1.340 log units

Regression R2	0.699
Classifier ROC AUC	0.954
Classifier Brier score	0.100
Conformal 95% half width	3.12 log units

Ordering is usable, individual values are not

Under repeated scaffold-grouped cross validation the regressor explains a real share of the variance across held out scaffolds (R2 = 0.70, RMSE 1.34 log units), and the classifier separates active from inactive well (ROC AUC = 0.95).

That does not make a predicted IC50 a quantity. The calibrated 95% conformal interval is about 3.1 log units in each direction, a factor of roughly 1,330 on IC50. Rank ordering within the generated set is defensible; a single predicted IC50 is not, and every pIC50 here is printed with its interval for that reason.

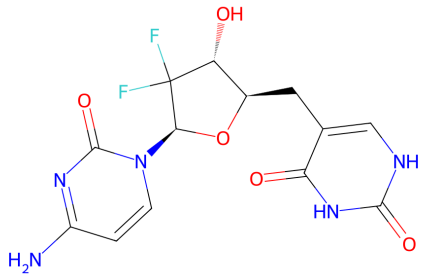
A note on method. These figures come from repeated grouped folds, not one split. A single scaffold split on 43 compounds leaves about ten test compounds and returns anything from R2 -0.09 to 0.82 depending only on the random seed, so a value from one split describes the split rather than the model.

2.4 Tissue simulation

Model	Two dimensional agent based PDAC microenvironment
Grid	36 x 28 cells over 800 x 600 micrometres
Fields	Oxygen and nutrient reaction diffusion, drug diffusion and uptake
Features	Hypovascularity, dense collagen matrix, sensitive and resistant clones
Comparison	Treated arm against an untreated arm with the same seed
Replicates	5 independent seeds per compound

Kinetics and microenvironment parameters are exploratory and have not been fitted to experimental pancreatic tumour data. The model's drug response is driven by the predicted IC50, so two compounds with the same predicted IC50 produce the same trajectory. Simulated growth inhibition is a property of this model, not a measurement.

Candidate 01: Compound-CFB9B6



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](Cc3c[nH]c(=O)[nH]c3=O)[C@@H](O)C2(F)F)c(=O)n1
InChIKey	SYMMKNWOSBOWTM-GTNGPMTGSA-N
Molecular formula	C13H13F2N5O5
Murcko scaffold	O=c1[nH]cc(CC2CCC(n3cccnc3=O)O2)c(=O)[nH]1
Origin	generation 4 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
357.3	-1.66	156.1	4	7	0.512	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	7.030	3.91 to 10.15	ordering signal
Implied IC50 (uM)	0.0933	spans orders of magnitude	not a quantity
Active probability	0.835	0.73 to 0.93	the stronger prediction
Applicability domain	0.64	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.737	0.000	0.512	0.916	0.643	0.572	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	9.38 % over 5 seeds
95% confidence interval	-6.75 % to 25.52 %
Interval excludes zero	no
Per seed values	17.1, -8.1, 26.2, 6.8, 4.9
Resistant clone fraction	0.345
Value used as selection objective	17.1 % on the search seed
Versus negative control	p = 0.076, Hedges g 1.17
Distinguishable, uncorrected	no
Distinguishable, Bonferroni corrected	no

Relation to known drugs

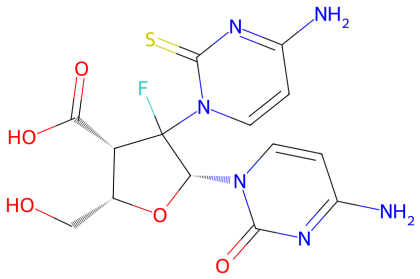
Nearest reference	Gemcitabine (Tanimoto 0.643)
Exact structural match	none in the local library
Library scope	28 curated reference drugs

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

Verdict

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

Candidate 02: Compound-5D822D



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](CO)[C@H](C(=O)O)C2(F)n2ccc(N)nc2=S)c(=O)n1
InChIKey	ZIEJAUJMRFDULL-SWVGTRJBSA-N
Molecular formula	C14H15FN6O5S
Murcko scaffold	O=c1ncccn1C1OCCC1n1cccnc1=S
Origin	generation 4 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
398.4	-0.75	171.5	4	9	0.480	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	7.135	4.01 to 10.26	ordering signal
Implied IC50 (uM)	0.0733	spans orders of magnitude	not a quantity
Active probability	0.838	0.74 to 0.94	the stronger prediction
Applicability domain	0.60	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.757	0.000	0.480	0.916	0.597	0.564	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	5.55 % over 5 seeds
95% confidence interval	-6.77 % to 17.87 %
Interval excludes zero	no
Per seed values	17.1, -8.1, 11.9, 6.8, 0.0
Resistant clone fraction	0.333
Value used as selection objective	17.1 % on the search seed
Versus negative control	p = 0.113, Hedges g 1.02
Distinguishable, uncorrected	no
Distinguishable, Bonferroni corrected	no

Relation to known drugs

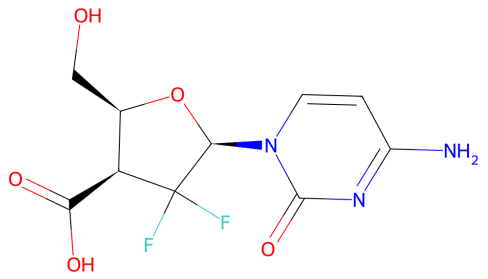
Nearest reference	Gemcitabine (Tanimoto 0.597)
Exact structural match	none in the local library
Library scope	28 curated reference drugs

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

Verdict

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

Candidate 03: Compound-FAFE93



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](CO)[C@H](C(=O)O)C2(F)F)c(=O)n1
InChIKey	HKQIZRDZFOYESW-UEYSZJFGSA-N
Molecular formula	C10H11F2N3O5
Murcko scaffold	O=c1ncccn1C1CCCO1
Origin	generation 4 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
291.2	-0.95	127.7	3	6	0.655	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	5.971	2.85 to 9.10	ordering signal
Implied IC50 (uM)	1.0685	spans orders of magnitude	not a quantity
Active probability	0.740	0.64 to 0.84	the stronger prediction
Applicability domain	0.77	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.493	0.000	0.655	0.916	0.766	0.544	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	1.63 % over 5 seeds
95% confidence interval	-8.93 % to 12.19 %
Interval excludes zero	no
Per seed values	0.0, -5.4, 9.5, 11.4, -7.3
Resistant clone fraction	0.312
Value used as selection objective	0.0 % on the search seed
Versus negative control	p = 0.248, Hedges g 0.71
Distinguishable, uncorrected	no
Distinguishable, Bonferroni corrected	no

Relation to known drugs

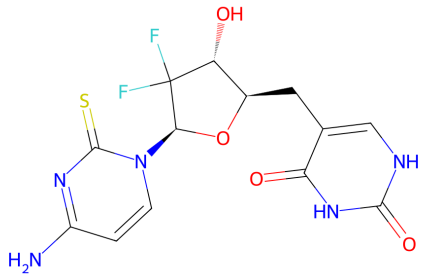
Nearest reference	Gemcitabine (Tanimoto 0.766)
Exact structural match	none in the local library
Library scope	28 curated reference drugs

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

Verdict

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

Candidate 04: Compound-C5F9E6



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](Cc3c[nH]c(=O)[nH]c3=O)[C@@H](O)C2(F)F)c(=S)n1
InChIKey	CONBZSYIZIVTIT-GTNGPMTGSA-N
Molecular formula	C13H13F2N5O4S
Murcko scaffold	O=c1[nH]cc(CC2CCC(n3cccnc3=S)O2)c(=O)[nH]1
Origin	generation 4 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
373.3	-0.29	139.0	4	7	0.542	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	6.967	3.84 to 10.09	ordering signal
Implied IC50 (uM)	0.1079	spans orders of magnitude	not a quantity
Active probability	0.858	0.76 to 0.96	the stronger prediction
Applicability domain	0.52	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.725	0.000	0.542	0.916	0.516	0.542	8.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	5.16 % over 5 seeds
95% confidence interval	-5.27 % to 15.58 %
Interval excludes zero	no
Per seed values	0.0, -2.7, 19.1, 4.5, 4.9
Resistant clone fraction	0.299
Value used as selection objective	0.0 % on the search seed
Versus negative control	p = 0.102, Hedges g 1.06
Distinguishable, uncorrected	no
Distinguishable, Bonferroni corrected	no

Relation to known drugs

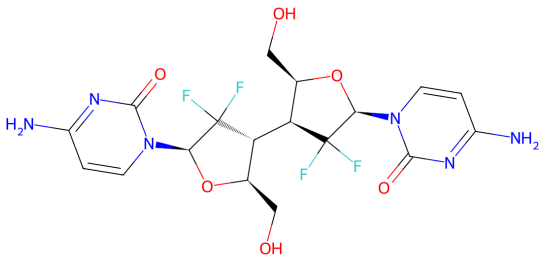
Nearest reference	Gemcitabine (Tanimoto 0.516)
Exact structural match	none in the local library
Library scope	28 curated reference drugs

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

Verdict

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

Candidate 05: Compound-32914B



Identity

Canonical SMILES	Nc1ccn([C@@H]2O[C@H](CO)[C@@H]([C@H]3[C@@H](CO)O[C@@H](n4ccc(N)nc4=O)C3(F)F)C2(F)F)c(=O)n1
InChIKey	LXFJNEGRNGGPAR-YIDOWPQCSA-N
Molecular formula	C18H20F4N6O6
Murcko scaffold	O=c1ncccn1C1CC(C2COC(n3cccnc3=O)C2)CO1
Origin	generation 4 variant, directed_optimization

Physicochemical profile

MW	cLogP	TPSA	HBD	HBA	QED	Lipinski
492.4	-1.05	180.7	4	10	0.376	passes

Predicted activity, with its uncertainty

Quantity	Value	95% interval	Reading
Predicted pIC50	7.103	3.98 to 10.23	ordering signal
Implied IC50 (uM)	0.0788	spans orders of magnitude	not a quantity
Active probability	0.810	0.71 to 0.91	the stronger prediction
Applicability domain	0.82	Tanimoto, nearest neighbour	inside domain

Objective scores and risk

Activity	Selectivity	QED	Low tox	Domain	Composite	Risk index
0.751	0.000	0.376	0.896	0.818	0.540	10.4 / 100

Simulated tissue response

Metric	Value
Tumour growth inhibition, mean	0.49 % over 5 seeds
95% confidence interval	-13.47 % to 14.45 %
Interval excludes zero	no
Per seed values	0.0, -13.5, 11.9, 11.4, -7.3
Resistant clone fraction	0.288
Value used as selection objective	0.0 % on the search seed
Versus negative control	p = 0.384, Hedges g 0.53
Distinguishable, uncorrected	no
Distinguishable, Bonferroni corrected	no

Relation to known drugs

Nearest reference	Gemcitabine (Tanimoto 0.818)
Exact structural match	none in the local library
Library scope	28 curated reference drugs

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

Verdict

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

3. What this run does not show

The value of a screening report lies as much in its boundaries as in its ranking. The following are not minor caveats; they determine what the numbers above can be used for.

No experimental evidence of any kind
No compound here has been synthesised. There is no cell assay, no animal model and no clinical data. Every number is the output of a model.

The tissue model is not calibrated to real tumours
Kinetics, diffusion constants, clone behaviour and drug response parameters are plausible but have not been fitted to experimental pancreatic tumour data. Simulated growth inhibition is a property of this software.

The potency model orders compounds, it does not quantify them
R2 of 0.70 under repeated grouped cross validation is real signal, but the conformal interval of about 3.1 log units still spans orders of magnitude in IC50. Use the ranking, not the numbers.

A small, narrow training set
43 curated compounds. Applicability domain is measured against that set, so high domain confidence means close to the training data, which also means structurally unadventurous.

Virtual time is not real time
Generations and exposure hours index the simulation. They do not correspond to chemical synthesis, biological timescales or clinical development.

Structural similarity is not efficacy and not novelty
Matching a known drug does not transfer its activity. Failing to match anything in a small local library does not make a structure new.

No ADME, formulation or synthesis assessment
Nothing here addresses whether these molecules can be made at reasonable cost, are stable, are orally available, or reach a pancreatic tumour in a living system. Dense desmoplastic stroma makes that last point a central problem in this disease.

Multiple comparisons
Five candidates were tested against one control arm. The per candidate threshold is uncorrected; a Bonferroni corrected threshold would be $p < 0.01$, and the results file records both.

Reference drugs and novel structures are not scored the same way
Gemcitabine's potency in the tissue model comes from curated experimental bioactivity. A newly generated structure has no measurement, so its potency comes from the structure-property model, which assigns micromolar values far weaker than the positive control's. At the dose used here that difference alone is close to sufficient to explain why no candidate matches the control. The comparison is therefore between a measurement and a prediction, not between two compounds, and failing to beat the control is weak evidence about the molecules.

The tissue objective is noisy and not independent
All arms in a generation share one seed, so selection sees a single draw of a quantity whose spread across seeds crosses zero. The objective also depends on the potency the pharmacology model assigns, so it is correlated with the activity objective rather than an independent check on it.

If you take one thing from this report
Pancreatic ductal adenocarcinoma has a five year survival rate in the low teens, and that is precisely why early computational output about it should be described carefully. This platform is a hypothesis generator. The appropriate next step for any compound listed here is scrutiny by a medicinal chemist, not optimism.

4. Reproducibility

4.1 Environment of this run

Platform	Windows-11-10.0.26200-SP0
Python	3.12.13
Compute backend	NumPy CPU float64
Device	host CPU
Field precision	float64
Search seed	20261027
Replicate seeds	20261027, 20261028, 20261029, 20261030, 20261031
Results hash	c2504b0fa0328c092e356ab2ca5b752c025eafaacfe0270c

4.2 Library versions

Package	Version
numpy	2.2.6
rdkit	2026.9.1
scikit-learn	1.9.1
scipy	1.18.1
torch	2.11.0+cu128

4.3 Source file hashes, SHA-256

File	SHA-256 (first 40 characters)
data\reference\pdac_reference_dataset.json	10de36cbf66cb3f74e2f79dd40a355847247b3bd
models\scientific\pdac_qsar_v1.joblib	3b310003b692ec09f7517a29b530e97e4108e78c
simulation\biology\diseases\pdac.py	38d708f39a7977c0d8dadd07e9c72bffe219306f
simulation\core\engine.py	4059a6fdea8603c90d4abe897475d3b48aaed4b0
simulation\discovery\pipeline.py	7aacc0fe78a202e3abcf096abed220e1ec83d0d0
simulation\discovery\workbench.py	a1bf63f8e409aa7e407d316beee8c21a06a20fcb
simulation\pharmacology\admet.py	064fe566b36565c07af097cb99a379543c6ddd4b

4.4 Reproducing this document

The platform is open source and so is this result. The reference dataset, the pipeline, the tissue models and the report generator are all in the repository. No model weights are required: the QSAR artifact is rebuilt from the committed dataset, and the optional language models were not used for any quantitative output.

```
python scripts/train_qsar.py

# Linux and macOS
EVOCURE_COMPUTE=cpu EVOCURE_DISABLE_LAYA_NEURAL=1 \
  python scripts/run_pdac_report_experiment.py

# Windows PowerShell
$env:EVOCURE_COMPUTE = "cpu"
$env:EVOCURE_DISABLE_LAYA_NEURAL = "1"
python scripts\run_pdac_report_experiment.py

python scripts/build_candidate_pdf.py
```

Forcing the CPU backend is what makes a run bit reproducible. GPU backends that support float64 agree to within 1e-12 on the substrate fields; float32 backends such as DirectML and Metal differ at around 1e-6, which is harmless for the conclusions and fatal for hash

equality. The backend used is recorded above and in the results file.

4.5 Licence and reuse

The platform is released under the MIT licence and the curated reference dataset under CC-BY 4.0. Results, including this report and the underlying JSON, are published so that the discoveries stay open: anyone may reproduce, check, contradict or build on them without permission.